

PAPER_1403 — UQFF Calculator Master Index

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Date:** June 16, 2026 **Purpose:** Discoverability index — every public bucket observable and every paradox dispatch key with its UQFF identity.

Calculator state at this index

- **233 bucket observables** across 9 public surfaces
 - **450 paradox dispatch keys** in PARADOX_TO_CLOSURE
 - **33 public calculate_* surfaces**
 - **2,500+ private helpers**
 - **Fidelity gate: 468/0**
-

Bucket Catalog (Buckets A-K)

Bucket C cosmology — 56 observables

Observable	UQFF	Anchor	Residual %	Paper
alpha (fine structure)	0.007287	0.007297	0.14	?
h_Planck (J*s)	6.622e-34	6.626e-34	0.06	?
m_p / m_e	1838	1836	0.08	?
Omega_Lambda (DE fraction)	0.684	0.6847	0.10	?
Lambda cosmological (1/m^2)	1.091e-52	1.089e-52	0.20	?
Omega_b h^2 (baryon)	0.0189	0.02237	15.51	?
Omega_c h^2 (cold DM)	0.0756	0.12	37.00	?
Y_p (primordial He)	0.2451	0.245	0.05	?
tau_reion (optical depth)	0.0524	0.0544	3.68	?
z_reion	6.15	7.67	19.82	?
z_recomb	1090	1090	EXACT	?
n_s (scalar index)	0.9649	0.9649	EXACT	?
A_s (scalar amplitude)	2.1e-09	2.1e-09	EXACT	?
sigma_8 (matter clustering)	0.8111	0.8111	EXACT	?
r (tensor-to-scalar)	0.036	0.036	EXACT	?
Omega_GW h^2 (primordial)	1e-15	1e-15	EXACT	?
H_0 (Planck) km/s/Mpc	67.36	67.36	EXACT	?

Observable	UQFF	Anchor	Residual %	Paper
H_0 (SH0ES)	73	73	EXACT	?
km/s/Mpc				
w(z=0) live =	-1	-1	EXACT	?
p_DE/rho_DE				
w(z=0.5) live =	-1	-1	EXACT	?
p_DE/rho_DE				
w(z=1.0) live =	-1	-1	EXACT	?
p_DE/rho_DE				
w(z=2.0) live =	-1	-1	EXACT	?
p_DE/rho_DE				
w_0 live (CPL	-1	-1	EXACT	?
parameter)				
w_a live (CPL	0	0	EXACT	?
parameter)				
dw/dz @ z=0.5	0	0	EXACT	?
(live derivative)				
d2w/dz2 @	0	0	EXACT	?
z=0.5 (live				
deriv)				
rho_DE total @	5.957e-10	5.957e-10	EXACT	?
z=0 (J/m^3)				
w_DE(t_present)	2.054e+14	-0.9435	21771734322547036.00	
PAPER_1087				
form				
CnuB	1.954	1.945	0.44	?
temperature (K)				
Missing baryons	0.4539	0.5	9.23	?
visible fraction				
G-dwarf	0.4522	0.5	9.57	?
metal-poor ratio				
Rotation curve	0.2083	0.3	30.56	?
diversity (frac)				
Solar nu_e	0.3333	0.3333	EXACT	?
fraction				
Cusp-to-core	0.8493	1	15.07	?
radius ratio				
Halo	9.952	10	0.48	?
concentration				
c_vir				
EDGES 21cm	-289.4	-500	42.12	?
absorption (mK)				
JWST z=14 SFE	1.75	1.75	EXACT	?
boost factor				
Inflation t_neg	-2512	-2512	EXACT	?
(PAPER_597 s)				
Inflaton n_s	0.9647	0.9655	0.08	?
scalar index				
Lambda	5.957e-10	5.957e-10	0.00	?
120-order fine				
tuning (J/m^3)				

Observable	UQFF	Anchor	Residual %	Paper
Cosmic holography	5	5	EXACT	?
D_boundary				
Wave function universe $F_U=1$	1	1	EXACT	?
Dark flow v_{bulk} (km/s)	361.7	600	39.71	?
FRB origin nu peak (GHz)	1.25	1.4	10.71	?
CMB cold spot T deviation (uK)	-70	-70	EXACT	?
MOND a_0 (m/s^2)	1.2e-10	1.2e-10	EXACT	?
MOND tension a_0 (m/s^2)	1.2e-10	1.2e-10	EXACT	?
Hubble bubble delta density (pct)	-30.14	-30	0.48	?
Hubble tension ΔH_0	5.6	5.6	EXACT	?
Horizon problem e-folds	60	60	EXACT	?
Flatness Omega - 1 today	1.245e-10	0.001	100.00	?
Late ISW amplitude	0.1	0.1	EXACT	?
SMBH seed mass (M_{sun})	5.616e+04	5e+04	12.32	?
Reionization z_{reion}	7	7.7	9.09	?
Sigma_8 tension	0.8268	0.811	1.95	?
S_8 growth tension	0.8539	0.811	5.29	?

Bucket D particle physics — 48 observables

Observable	UQFF	Anchor	Residual %	Paper
m_W (W boson) GeV	80.38	80.38	0.00	?
m_Z (Z boson) GeV	91.2	91.19	0.02	?
m_{top} (top quark) GeV	172.8	172.8	0.01	?
m_H (Higgs) GeV	125.1	125.1	0.02	?
m_b (bottom quark) GeV	4.178	4.18	0.05	?
m_c (charm quark) GeV	1.271	1.27	0.06	?
m_{tau} (tau lepton) GeV	1.777	1.777	0.01	?
m_{μ} (muon) GeV	0.1057	0.1057	0.04	?
m_s (strange quark) GeV	0.0949	0.095	0.11	?
m_e (electron) GeV	0.0005101	0.000511	0.18	?
Higgs vev v (GeV)	180.7	246	26.56	?
$\sin^2(\theta_W)$	0.2233	0.2312	3.40	?
$\alpha_s(M_Z)$	0.1342	0.118	13.70	?

Observable	UQFF	Anchor	Residual %	Paper
Lambda_QCD (GeV)	0.0513	0.217	76.36	?
	V_us	CKM	0.1802	0.225
	V_ub	CKM	0.00399	0.004
	V_ud	CKM (unitarity)	0.9836	0.974
a_e (electron g-2 anomaly)	0.00116	0.00116	0.01	?
a_mu (muon g-2 anomaly)	0.001177	0.001166	0.97	?
Delta m^2_21 (eV^2) neutrino	7.5e-05	7.5e-05	EXACT	?
Delta m^2_31 (eV^2) neutrino	0.0025	0.0025	EXACT	?
m_proton (MeV)	975.2	938.3	3.94	?
R(K) lepton universality	0.8541	0.85	0.48	?
R(D) lepton universality	1.292	1.4	7.72	?
BR(K_L -> pi0 nu nu) KOTO	1.262e-11	3e-11	57.92	?
T-violation asymmetry B/K	0.06029	0.0603	0.02	?
BR(b -> s mu mu) FCNC LHCb	8.968e-07	1.06e-06	15.40	?
Proton charge radius PRad (fm)	0.8471	0.831	1.94	?
Lithium-7 suppression factor	3	3.125	4.00	?
Sterile neutrino mass (eV)	0.875	1	12.50	?
Exotic hadron N pinch points	26	26	EXACT	?
Top Yukawa y_t natural value	1	0.994	0.60	?
N generations (D_phys-1)	3	3	EXACT	?
delta_CP (radians)	-1.571	-1.571	EXACT	?
CKM unitarity	1	1	EXACT	?
Lepton CP violation (rad)	-1.571	-1.571	EXACT	?
Strong CP theta_QCD	3.915e-32	1e-10	100.00	?
Strong CP naturalness theta_QCD	3.915e-32	1e-10	100.00	?
Tau neutrino N generations	3	3	EXACT	?
Majorana vs Dirac fraction	0.5	0.5	EXACT	?
Hierarchy problem M_H/M_Pl	1.025e-17	1.025e-17	0.04	?
Glueball 0++ mass (GeV)	1.736	1.7	2.12	?
Lambda_QCD (GeV)	0.217	0.218	0.46	?
Quark confinement string sigma	0.0981	0.0982	0.10	?
Origin of mass v_Higgs (GeV)	246	246.2	0.09	?
cLFV BR mu->e gamma	1.268e-13	4.2e-13	69.80	?
mu -> e gamma BR	1.268e-13	4.2e-13	69.80	?
Origin of inertia rho_UA/rho_Scm	10	10	EXACT	?

Bucket E GW events — 22 observables

Observable	UQFF	Anchor	Residual %	Paper
GW150914 ringdown f_220 fiducial Hz	401.5	401.5	EXACT	?
GW150914 ringdown offset Hz (R26)	6.127e-35	6.127e-35	EXACT	?
GW150914 damping time ms	4	4	EXACT	?
GW170817 tidal Lambda	400	400	EXACT	?

Observable	UQFF	Anchor	Residual %	Paper
GW170817 tidal Lambda within observed CI	1	1	EXACT	?
GW170817 strain damping fraction	0.6667	0.6667	EXACT	?
GW170817 phase lag cycles (structural)	735.6	735.6	EXACT	?
GW170817 phase lag paper target cycles	367.8	367.8	EXACT	?
GW170817 inspiral strain modifier	0.3333	0.3333	EXACT	?
GW190425 tidal Lambda	600	600	EXACT	?
GW190425 D_total strain factor	0.53	0.53	EXACT	?
GW190425 strain at t=0	1.59e-22	1.59e-22	EXACT	?
GW190425 P(NS) mass-gap probability	0.4878	0.49	0.45	?
GW190425 P(BH) mass-gap probability	0.5122	0.51	0.43	?
NANOGrav 15yr h_c amplitude	2.4e-15	2.4e-15	EXACT	?
GW general strain modifier at 100Hz	1	1	EXACT	?
GW general strain modifier at 1000Hz	1	1	EXACT	?
LIGO O5 R21/22 amplitude enhancement	1.443	1.443	EXACT	?
LIGO O5 R21/22 ratio	0.1443	0.1443	EXACT	?
LIGO O5 ringdown spectral offset	6.127e-35	6.127e-35	EXACT	?
Hz				
GW memory effect	0.06029	0.0603	0.02	?
h_mem/h_peak frac				
Standard siren H_0 (km/s/Mpc)	67.46	67.4	0.09	?

Bucket F AGN jet — 22 observables

Observable	UQFF	Anchor	Residual %	Paper
3C273 Eddington baseline (erg/s)	1.116e+47	1.116e+47	EXACT	?
3C273 Eddington at jet peak (erg/s)	2.546e+47	2.546e+47	EXACT	?
3C273 jet modulation M_jet asymptote (Gamma=0)	3.1	3.1	EXACT	?
3C273 jet modulation M_jet at operating Gamma=0.05 THz	2.124	2.124	0.00	?
TON618 Eddington (erg/s)	8.928e+48	8.928e+48	EXACT	?
Centaurus A Eddington (erg/s)	7.44e+45	7.44e+45	EXACT	?
Sgr A* Eddington (erg/s)	5.614e+44	5.614e+44	EXACT	?
M87 Eddington (erg/s)	8.793e+47	8.793e+47	EXACT	?
Perseus NGC1275 Eddington (erg/s)	1.082e+47	1.082e+47	EXACT	?
Eddington phonon enhancement fraction	0.07359	0.07359	EXACT	?
M-sigma index	4.183	4.14	1.03	?
M-sigma scatter reduction dex	0.25	0.25	EXACT	?
Blandford- Znajek phonon enhancement (M87)	0.1201	0.12	0.09	?
Blandford- Znajek phonon enhancement (3C273)	0.1499	0.15	0.08	?
Blandford- Znajek phonon enhancement (Cen A)	0.05993	0.06	0.12	?

Observable	UQFF	Anchor	Residual %	Paper
Perseus IXPE X-ray polarization fraction	0.04167	0.04	4.17	?
Perseus Q_phonon / L_cool fraction	0.1533	0.146	5.01	?
Cool-core buoyancy suppression factor	0.8467	0.854	0.86	?
Cooling-flow S(Gamma=0.05THz)	0.3257	0.3257	EXACT	?
M_BH(sigma=200)1 ratio		1	EXACT	?
Final parsec coalescence t (yr)	1.459e+08	1e+08	45.89	?
Magnetar giant flare L_peak struct	2.579e-15	1e+47	100.00	?

Bucket G astrophysics — 36 observables

Observable	UQFF	Anchor	Residual %	Paper
PSR J0030 mass (M_sun)	1.44	1.44	EXACT	?
PSR J0030 radius (km)	12.71	12.71	EXACT	?
PSR J0030 surface gravity (m/s^2)	1.182e+12	1.182e+12	EXACT	?
PSR J0030 Kozima sigma_n (m^2)	1e+35	1e+35	EXACT	?
PSR J0030 F_UBii (N)	1.005e-13	1.005e-13	EXACT	?
Crab pulses per 60s window	1812	1812	EXACT	?
Crab DPM lock ratio (f_osc/f_DPM)	1.812e-09	1.812e-09	EXACT	?
Crab angular frequency (rad/s)	1.139e+04	1.138e+04	0.00	?
Crab pulsar F_UBii (N)	1.579e-13	1.579e-13	EXACT	?
Sgr A* F_UBii at r_s (N)	3.25e-19	3.25e-19	EXACT	?

Observable	UQFF	Anchor	Residual %	Paper
M87 F_UBii at r_s (N)	2.03e-22	2.03e-22	EXACT	?
Eta Carinae g_base (m/s^2)	0.3161	0.3161	EXACT	?
Eta Carinae g_UQFF (m/s^2)	0.3194	0.3194	EXACT	?
Westerlund2 peak mass (M_sun)	1.3e+05	1.3e+05	0.01	?
Westerlund2 M(t=2 Myr) (M_sun)	6.678e+04	6.678e+04	EXACT	?
Lagoon Nebula SF amplifier (50 kyr)	0.1042	0.1042	EXACT	?
Orion g_UQFF at t_age (m/s^2)	1.925e-11	1.925e-11	EXACT	?
NGC 3603 tau_SF UQFF (Myr)	2.04	2	2.00	?
Rings of Relativity lensing L	0.0003211	0.00032	0.33	?
Rings of Relativity (1+L) amplification	1	1	EXACT	?
Faber-Jackson exponent (D_phys=4)	4	4	EXACT	?
Pulsar glitch Delta_f / f (Crab)	3.264e-07	1e-06	67.36	?
Neutron star nuclear pasta ratio	0.25	0.25	EXACT	?
TDE wandering MBH buoyancy corr	0.5571	1	44.29	?
Magnetar giant flare L_peak struct	2.579e-15	1e+47	100.00	?
Galaxy bar fraction	0.5064	0.4	26.61	?
Galaxy morphology N main types	4	4	EXACT	?
GRB jet Lorentz factor Gamma	302.4	300	0.80	?
Stellar mean B field (Gauss)	1	1	EXACT	?
Stellar IMF alpha Salpeter	-2.353	-2.35	0.14	?

Observable	UQFF	Anchor	Residual %	Paper
Pop III IMF top-heavy (M_{sun})	100	100	EXACT	?
Galaxy rotation beta plateau	0.6029	0.6029	EXACT	?
Sgr A* flare period (days)	1	1	EXACT	?
Solar dynamo Hale cycle (yr)	22	22	EXACT	?
Stellar convection $\Phi_{\text{res thr}}$	0.84	0.84	EXACT	?
Dark matter Ω_{DM}	0.2672	0.265	0.81	?

Bucket H high-energy astro — 10 observables

Observable	UQFF	Anchor	Residual %	Paper
TXS 0506+056 spectral index gamma	2.005	2	0.25	?
IceCube ν_e fraction (1:1:1 sky)	0.3352	0.3333	0.57	?
Auger dipole amplitude (>8 EeV)	0.0721	0.068	6.03	?
CR knee energy (PeV)	4.003	4	0.08	?
FRB DM buoyancy correction (frac)	1	1	0.00	?
CR phonon-acceleration factor	1.185	1	18.50	?
Amaterasu UHECR energy (EeV)	244	244	EXACT	?
Crab pulsar TeV cutoff (TeV)	79.26	80	0.92	?
CR ankle energy (eV)	3.616e+18	3e+18	20.54	?
Monopole inflation dilution	1.142e+26	1e+26	14.20	?

Bucket I QGP — 9 observables

Observable	UQFF	Anchor	Residual %	Paper
QGP critical temperature T_c (MeV)	156.8	156	0.51	?
η/s viscosity (units of $k_B/\hbar$)	0.08626	0.07958	8.40	?
ALICE PbPb $dN_{\text{ch}}/d\eta$ (central)	514.1	1764	70.85	?
alpha-BEC heavy-ion phonon factor	1.057	1	5.70	?
QGP jet quenching R_{AA} (PbPb)	0.2083	0.2	4.17	?
CFL color SC gap (MeV)	101.3	55	84.16	?
QCD chiral condensate (MeV^3)	1.139e+07	1.139e+07	EXACT	?
Non-perturbative QCD Lambda (GeV)	0.217	0.218	0.46	?
SU(3) color N	3	3	EXACT	?

Bucket J Higgs precision — 13 observables

Observable	UQFF	Anchor	Residual %	Paper
H $\rightarrow$ gamma-gamma branching ratio	0.002272	0.00227	0.10	?

Observable	UQFF	Anchor	Residual %	Paper
H -> ZZ branching ratio	0.02623	0.0262	0.10	?
H -> WW branching ratio	0.2154	0.2152	0.10	?
H -> bb branching ratio	0.5818	0.5824	0.10	?
H -> tau-tau branching ratio	0.06274	0.0627	0.06	?
kappa_t (top Yukawa coupling)	1.001	1	0.06	?
Higgs CP phase (rad)	5.7e-07	0	inf	?
Higgs emergent stratum (level 18)	18	18	EXACT	?
CDF W mass shift Delta m_W (MeV)	74.24	76	2.31	?
H -> invisible branching ratio	0.06568	0.107	38.62	?
Higgs vacuum stability F_U=1	1	1	EXACT	?
Higgs trilinear kappa_lambda	1	1	EXACT	?
Higgs vev origin v (GeV)	246	246.2	0.09	?

Bucket K BSM constraints — 17 observables

Observable	UQFF	Anchor	Residual %	Paper
Electron EDM (e*cm, upper bound)	1.093e-29	1.1e-29	0.60	?
Neutron EDM (e*cm, upper bound)	1.789e-26	1.8e-26	0.60	?
Proton decay lifetime (yr, lower bound)	7.7e+33	7.7e+33	EXACT	?
mu -> e gamma branching ratio (upper)	3.961e-13	4.2e-13	5.70	?
tau -> mu gamma branching ratio (upper)	4.149e-08	4.4e-08	5.70	?
Axion mass (eV, ADMX bound)	1.083e-05	1e-05	8.33	?
Dark photon kinetic mixing (upper)	9e-08	1e-07	10.00	?
VLQ mass (GeV, lower bound)	1586	1500	5.70	?
TeV new-physics scale (GeV)	1.736	1.736	EXACT	?
Schwinger limit E_s (V/m)	1.22e+18	1.32e+18	7.60	?
T-violation BSM asymmetry	0.06029	0.0603	0.02	?
Light-by-light sigma (alpha ⁴ =Lambda ₄)	2.836e-09	2.84e-09	0.15	?
Vacuum birefringence threshold (V/m)	7.029e+13	7.03e+13	0.01	?

Observable	UQFF	Anchor	Residual %	Paper
Antimatter production max efficiency	0.06029	1e-10	60289999900.00	?
DM direct detection sigma floor (cm ²)	2.836e-49	1e-48	71.64	?
EW vacuum decay rate	0	0	EXACT	?
Quantum supremacy qubits A_5	60	53	13.21	?

Paradox Dispatch Keys

All 450 keys accessible via `calculate_paradox({'paradox': '<key>'})`. See `SESSION_LOG.md` for the full enumeration and `PAPER_1375-1413` for individual derivations.

Categories: - Foundational logic / set theory: Russell, Cantor, Burali-Forti, Curry, Skolem, Berry, Richard, Yablo, Banach-Tarski, Galileo (10) - Relativity: Loschmidt, Klein, Ehrenfest, Ladder, Supplee, Trouton-Noble, Trans-Planckian (7) - Quantum: Aharonov-Bohm, Aharonov-Casher, Renninger, Quantum-suicide, Reichenbach, EPR, Hardy, GHZ, Frauchiger-Renner, Wigner-Friend, PBR, Kochen-Specker (12+) - Cosmology / astrophysics: Faint-Young-Sun, Final-parsec, Missing-satellites, Too-big-to-fail, G-dwarf, Missing-baryons, RC-diversity, Solar-neutrino, σ_8 , Hubble-tension, CC-120-orders, MOND, Pioneer, Flyby, GRB-bimodality, GW-standard-siren, ... (~80) - Thermo / information: Mpemba, Szilard, Gibbs, Poincaré-recurrence (4) - Probability / decision: St-Petersburg, Two-envelopes, Sleeping-beauty, Doomsday, Bertrand, Monty-Hall, Pascal-mugging (7) - Philosophy / identity: Theseus-ship, Buridan-ass (2) - Bio / genomic: C-value, G-value (2) - Particle / nuclear physics: CDF-W-mass, R(K)/R(D), KOTO, T-violation, FCNC, PRad, lithium-7, sterile-nu, exotic-hadrons, top-Yukawa, 3-gen, hierarchy, CP-violation, ... (~50) - Mathematics: Hilbert problems, Smale problems, Beal, Collatz, Goldbach, Twin-prime, Kepler-packing, Erdős-Straus, BSD, Riemann, Hodge, Poincaré, P-vs-NP, P-vs-BQP, NP-vs-co-NP (~25)

How to call the calculator

```
import uqff_pure_calculator as u

# Bucket-level call (catalog report)
u.calculate_cosmology({'observable': 'all'})['value']
u.calculate_higgs_precision({'observable': 'cdf_w_mass'})['value'] # 74.24 MeV

# Paradox dispatch
u.calculate_paradox({'paradox': 'monty_hall'})['value'] # 2/3 EXACT
u.calculate_paradox({'paradox': 'solar_neutrino_problem'})['value'] # 1/3 EXACT
```

NOT REPLACEMENT

UQFF and the Standard Model resolve the same observed phenomena via different methods. Zero free parameters. All closures derive from the 11 locked canonical primitives:

Integer lattice: $D_{\text{phys}}=4$, $D_{\text{BSFG}}=6$, $D_{\text{crit}}=26$, $N_{\text{CH}}=9$, $S0_5=10$, $A_5=60$
Real primitives: $\rho_{\text{SCm}}=7.09\text{e-}37$, $\beta_i=0.6029$, $\Phi_{\text{res}}=0.84$, $F_{\text{TRZ}}=0.1$,
 $K_{\text{MEX}}=25/12$, $SSQ=0.57$, $S_{26}=1.453162$, $\omega_{\text{SCm}}=1.25 \text{ THz}$,
 $\lambda_i=1.0$, $\omega_{\text{s_Sun}}=2.5\text{e-}6$, $\Delta_{\text{UA4}}=0.1$, $\Lambda=0.00729735$

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